

Due tuesday, February 12th, noon.

This is the first problem set. Please present clean drawings/sketches wherever relevant, and show all calculations up to reasonable detail.

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1. (a) [**1+1 pts.**] Look up and report the values of the *permittivity of free space*, ϵ_0 , and the *permeability of free space*, μ_0 , in SI units. Include the units of each quantity.

(b) [**2 pts.**] Look up and report Coulomb's law in SI units and in CGS units.

2. We will ignore the z direction for this problem. Consider the following points on the x - y plane: the point A with coordinates $(0, a)$, the point B with coordinates $(0, -a)$, and the point P with coordinates $(r, 0)$.

(a) [**5 pts.**] A particle at point P feels two forces, both of equal magnitude F_0 . One force points in the direction from A towards P (along the direction AP), and the other acts towards B (along the direction PB). Find the total force acting on the particle.
Reminder: Finding a vector means finding both its magnitude and its direction!!

(b) [**5 pts.**] Imagine two vectors, each having the same magnitude E_0 , acting on point P . One points from P towards A (along the direction PA), and the other points from P towards B (along the direction PB). Find the result of adding the two vectors.

3. [**8 pts.**] A thin glass rod lies along the x -axis, with one end at the origin and one end at the point $(L, 0, 0)$. The rod is charged non-uniformly. The linear charge density is $\lambda = 4\gamma x^3$. What is the total charge on the rod?
Note: the linear charge density is the charge per unit length. Here γ is a constant.

You will have to divide the rod into infinitesimally small pieces, and 'add up' the charge on each of these pieces. Adding up will mean performing an integration.

4. [9 pts.] A flat square plate lies parallel to the x - y plane and is charged with a variable surface charge density, $\sigma = 2\gamma(xy + x^2)$. (Here γ is a constant.) The four corners of the square plate are at the points $(0, 0, z_0)$, $(L, 0, z_0)$, $(0, L, z_0)$, (L, L, z_0) . What is the total charge on the plate?

Note: the surface charge density is the charge per unit area. In this case a two-dimensional integration will be required.

5. [4 pts.] The electric field in some region is given by

$$\mathbf{E} = \frac{V_0}{d} (5xy\hat{i} + 2z^2\hat{j})$$

where $\hat{i}$ and $\hat{j}$ are unit vectors in the x and y directions, and V_0 and d are positive constants. Find the force experienced by a charge q that is placed at the point $(x, y, z) = (d, -d, d)$.

6. [7 pts.] The closed curve C lies in the x - y plane and encloses area α . Using Stokes' theorem, calculate the line integral of the vector field

$$\mathbf{T} = -2y\hat{i} + 2x\hat{j}$$

around the curve C .

7. Here L and d are positive distances, with $d > L$. Also, Q and q are positive charges.

- (a) [5 pts.] Five point charges are placed on the x -axis, at the points with x -coordinates $L/5$, $2L/5$, $3L/5$, $4L/5$ and L . Each of these carry the charge $Q/5$.

A point charge q is placed, also on the x -axis, at the point $(d, 0, 0)$. Find the total force experienced by the charge q .

- (b) [3 pts.] Consider the limit $d \gg L$. Simplify your expression for this limit, and explain your result physically.